Exercise 5: Mapping using 2D Lidar and ICP

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Task 1

Background Data

Based on the first 3 steps of the exercise, I have the following initial matrices and vectors:

$$H = \begin{bmatrix} 6.4 & 4.8 \\ 0 & 0 \end{bmatrix}, \quad \bar{s} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \bar{m} = \begin{bmatrix} 2.2 \\ 2.4 \end{bmatrix}$$

Step 4: Generate a singular value decomposition (SVD) of the covariance matrix

To find the optimal rotation, I first need to decompose the covariance matrix H into three components such that $H = U\Sigma V^T$. As explicitly permitted by the exercise instructions, I computed this Singular Value Decomposition using `numpy.linalg.svd`, which yields the unitary matrix U, the diagonal matrix of singular values Σ, and the unitary matrix V:

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 8 & 0 \\ 0 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$$

Step 5: Compute the determinant of the unitary matrices

Next, I compute the determinant of the product of the unitary matrices, $d = \det(VU^T)$. This step is conceptually crucial because it helps identify whether the resulting geometric transformation involves an invalid reflection. If the determinant is -1, it indicates a reflection; if it is $+1$, it is a proper spatial rotation.

First, I calculate the matrix product VU^T:

$$VU^T = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$$

Now, I compute its determinant:

$$d = \det(VU^T) = (0.8 \cdot 0.8) - (-0.6 \cdot 0.6) = 0.64 + 0.36 = 1$$

Since $d = 1$, I can be confident that the transformation is a valid rotation and no reflection correction is required.

Step 6: Compute the optimal rotation matrix

Now, I can compute the optimal rotation matrix R. To mathematically guarantee that R is a proper rotation matrix in $SO(2)$ (meaning $\det(R) = 1$), I include the determinant d as an adjustment factor within the diagonal matrix:

$$R = V \begin{bmatrix} 1 & 0 \\ 0 & d \end{bmatrix} U^T$$

Substituting the known matrices:

$$R = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$$

Step 7: Compute the translation vector

With the optimal rotation R successfully found, I need to calculate the optimal translation vector t. This vector aligns the centers of the two point clouds by shifting the rotated centroid of the scan points ($\bar{s}$) directly onto the centroid of the map points ($\bar{m}$):

$$t = \bar{m} - R\bar{s}$$

First, I apply the rotation to the scan centroid:

$$R\bar{s} = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.8(1) - 0.6(1) \\ 0.6(1) + 0.8(1) \end{bmatrix} = \begin{bmatrix} 0.2 \\ 1.4 \end{bmatrix}$$

Then, I subtract this result from the map centroid to find the final translation:

$$t = \begin{bmatrix} 2.2 \\ 2.4 \end{bmatrix} - \begin{bmatrix} 0.2 \\ 1.4 \end{bmatrix} = \begin{bmatrix} 2.0 \\ 1.0 \end{bmatrix}$$

Step 8: Construct the homogeneous transformation matrix

Finally, I construct the homogeneous transformation matrix $T \in \mathbb{R}^{3\times3}$. This step is practically necessary because it neatly packages both the rotation R and the translation t into a single square matrix. By doing this, any 2D scan point (appended with a 1 to make it a 3D vector) can be fully transformed into the map frame using a single matrix-vector multiplication.

$$T = \begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix}$$

Plugging in the previously calculated R and t:

$$T = \begin{bmatrix} 0.8 & -0.6 & 2.0 \\ 0.6 & 0.8 & 1.0 \\ 0 & 0 & 1 \end{bmatrix}$$